Release and rollback

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete release-dependency and gate record.

Keep versions, current data compatibility, required gates, failed evidence and reviewed recovery conditions together.

Keep one habit: Recall → W1–W3/W5 → independent prompt/corpus case → release manifest → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Release records bind code, configuration, data schema and evaluation **evidence**. **Rollback** requires **compatibility** with the current stored data.

- Version v2 writes schema 2. Version v1 reads only schema 1. A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Deployment environments

> **My remembered idea, example and correction**

Caching and stale information

> **My remembered idea, example and correction**

Untrusted input boundaries

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Release records bind code, configuration, schema, prompt, corpus and evaluation evidence. Main A manifest uses v1/cfg 1/schema 1/p1/c1/eval-A; B uses v2/cfg 2/schema 2/p2/c2/eval-B. Reader v1 supports schema 1 only; reader v2 supports 1 and 2. Current stored schemas [1, 2, 2] therefore block a code-only v1 rollback.

  A compatible-reader check alone does not establish complete recovery readiness; a migration or adapter needs its own tested plan. Separate picture: A 8/10 = 80%, B 9/10 = 90%, a 10-percentage-point rise on supplied cases. B still fails a mandatory privacy gate. A fresh reserved privacy pass adds scoped evidence but does not satisfy every gate.

  The code supplies a fictional 80% quality threshold and explicit quality/privacy/compatibility/recovery Booleans; any failure blocks, all passing permits review only. Keep original failure and new repair evidence versioned. Code rollback does not automatically undo data changes or external effects.

  The program makes pure decisions over fictional manifests and leaves its schema inventory unchanged; it performs no deployment, migration or restoration.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Release records bind code, configuration, schema, prompt, corpus and evaluation evidence. Main A manifest uses v1/cfg 1/schema 1/p1/c1/eval-A; B uses v2/cfg 2/schema 2/p2/c2/eval-B. Reader v1 supports schema 1 only; reader v2 supports 1 and 2. Current stored schemas [1, 2, 2] therefore block a code-only v1 rollback.

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  The program makes pure decisions over fictional manifests and leaves its schema inventory unchanged; it performs no deployment, migration or restoration.

- After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Release records bind code, configuration, schema, prompt, corpus and evaluation evidence. Main A manifest uses v1/cfg 1/schema 1/p1/c1/eval-A; B uses v2/cfg 2/schema 2/p2/c2/eval-B. Reader v1 supports schema 1 only; reader v2 supports 1 and 2. Current stored schemas [1, 2, 2] therefore block a code-only v1 rollback.

  A compatible-reader check alone does not establish complete recovery readiness; a migration or adapter needs its own tested plan. Separate picture: A 8/10 = 80%, B 9/10 = 90%, a 10-percentage-point rise on supplied cases. B still fails a mandatory privacy gate. A fresh reserved privacy pass adds scoped evidence but does not satisfy every gate.

  The code supplies a fictional 80% quality threshold and explicit quality/privacy/compatibility/recovery Booleans; any failure blocks, all passing permits review only. Keep original failure and new repair evidence versioned. Code rollback does not automatically undo data changes or external effects.

  The program makes pure decisions over fictional manifests and leaves its schema inventory unchanged; it performs no deployment, migration or restoration.

- A report omits this qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Release records bind code, configuration, schema, prompt, corpus and evaluation evidence. Main A manifest uses v1/cfg 1/schema 1/p1/c1/eval-A; B uses v2/cfg 2/schema 2/p2/c2/eval-B. Reader v1 supports schema 1 only; reader v2 supports 1 and 2. Current stored schemas [1, 2, 2] therefore block a code-only v1 rollback.

  A compatible-reader check alone does not establish complete recovery readiness; a migration or adapter needs its own tested plan. Separate picture: A 8/10 = 80%, B 9/10 = 90%, a 10-percentage-point rise on supplied cases. B still fails a mandatory privacy gate. A fresh reserved privacy pass adds scoped evidence but does not satisfy every gate.

  The code supplies a fictional 80% quality threshold and explicit quality/privacy/compatibility/recovery Booleans; any failure blocks, all passing permits review only. Keep original failure and new repair evidence versioned. Code rollback does not automatically undo data changes or external effects.

  The program makes pure decisions over fictional manifests and leaves its schema inventory unchanged; it performs no deployment, migration or restoration.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled release and **rollback evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of release and **rollback**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Release and **rollback**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Release and **rollback**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

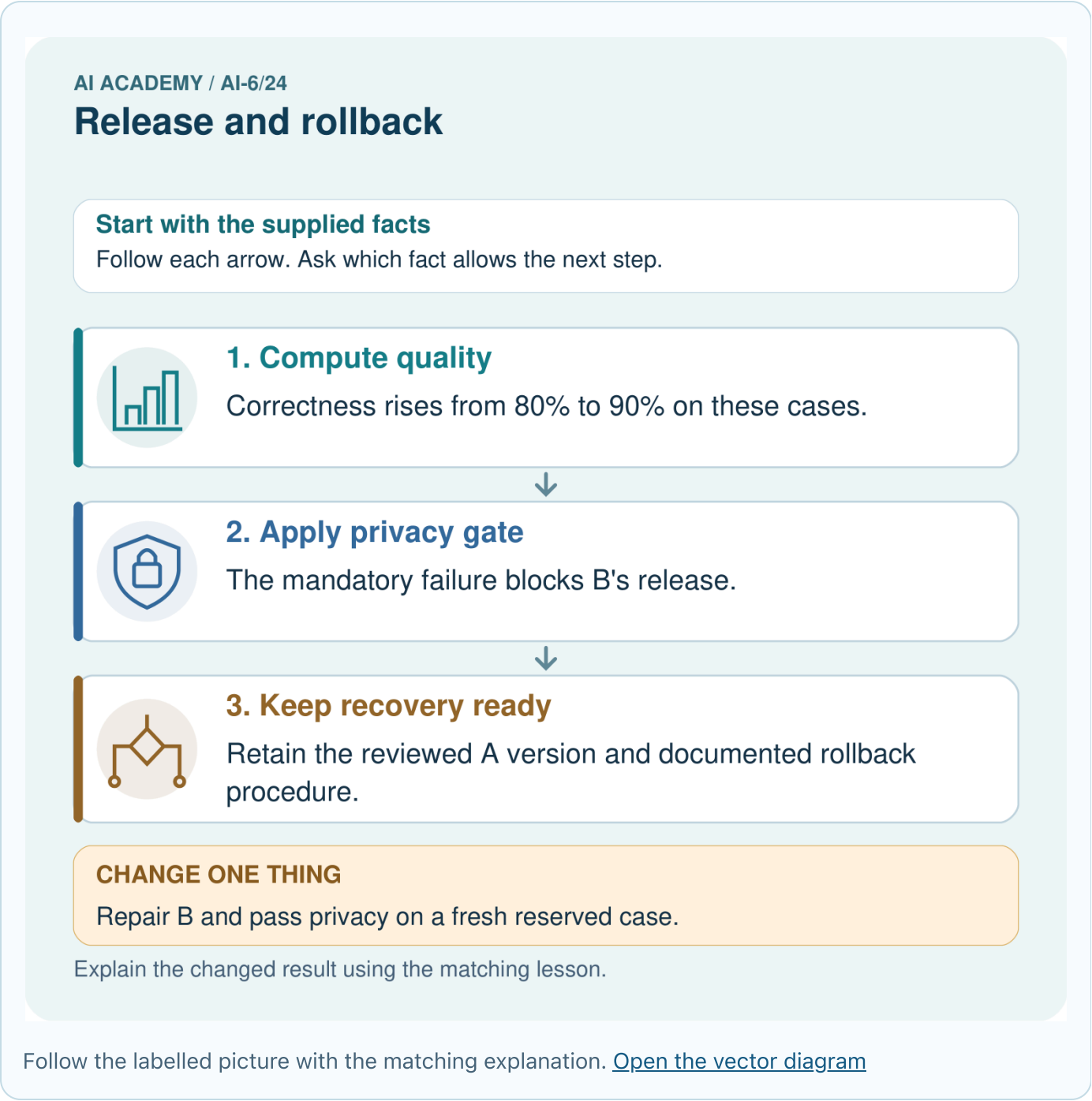
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.
- A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.
- Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.
- Which labelled step matches this detail? Retain the reviewed A version and documented **rollback** procedure.
- 1. Keep recovery ready
- 2. Apply privacy gate

- 3. Compute quality

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep recovery ready Retain the reviewed A version and documented **rollback** procedure. Release criteria combine benefits and mandatory conditions.

Rollback requires known versions and compatible data, not an informal promise.

Changed-case question and optional hint

- Changed-case practice: Repair B and pass privacy on a fresh reserved case. Which conclusion is supported by the supplied facts?

- 1. B may enter the remaining gates; one repair does not approve everything.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: B may enter the remaining gates; one repair does not approve everything. All required release conditions still apply.

Repair and retain the record

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version -5 request return, and what permits a later version -5 cache hit?. Return to this week and explain what you changed.

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Release and rollback: the case



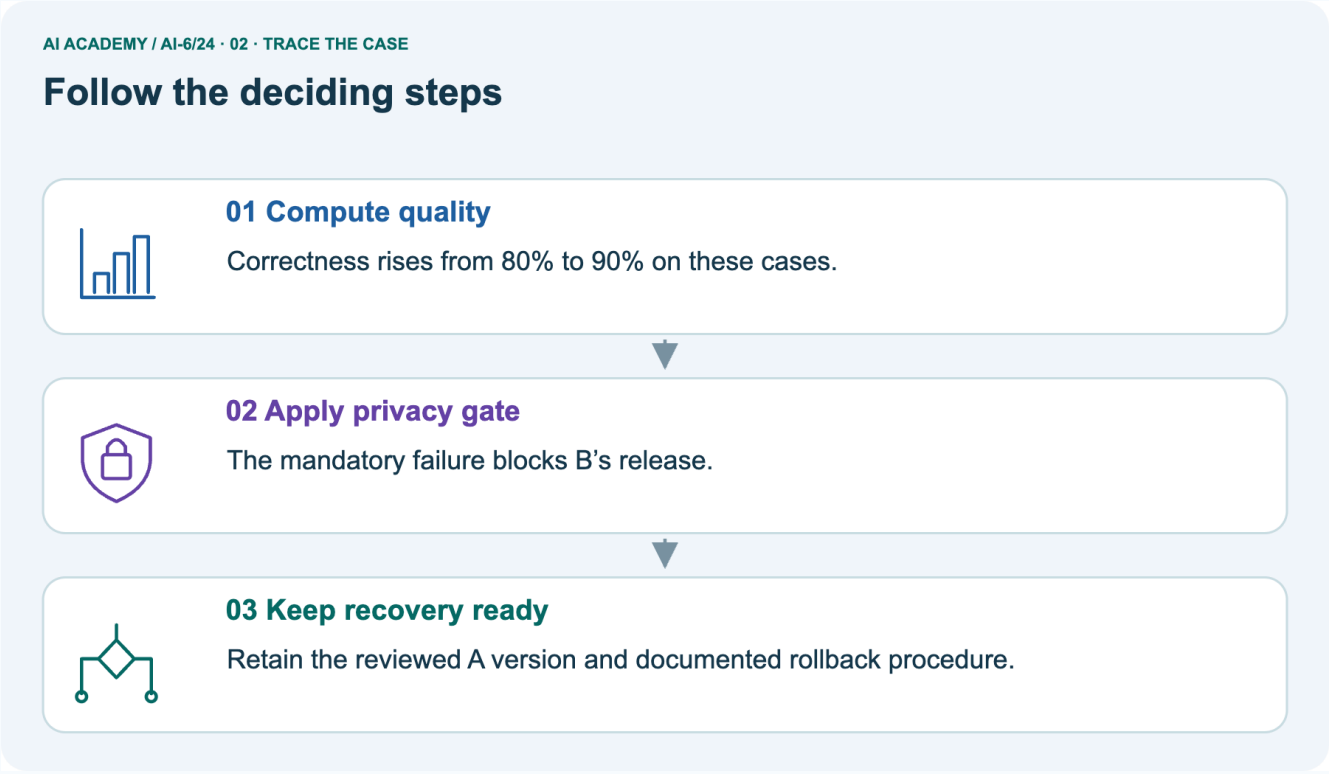
Follow the labelled picture with the matching explanation. Open the vector diagram

A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory. Supplied facts: A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



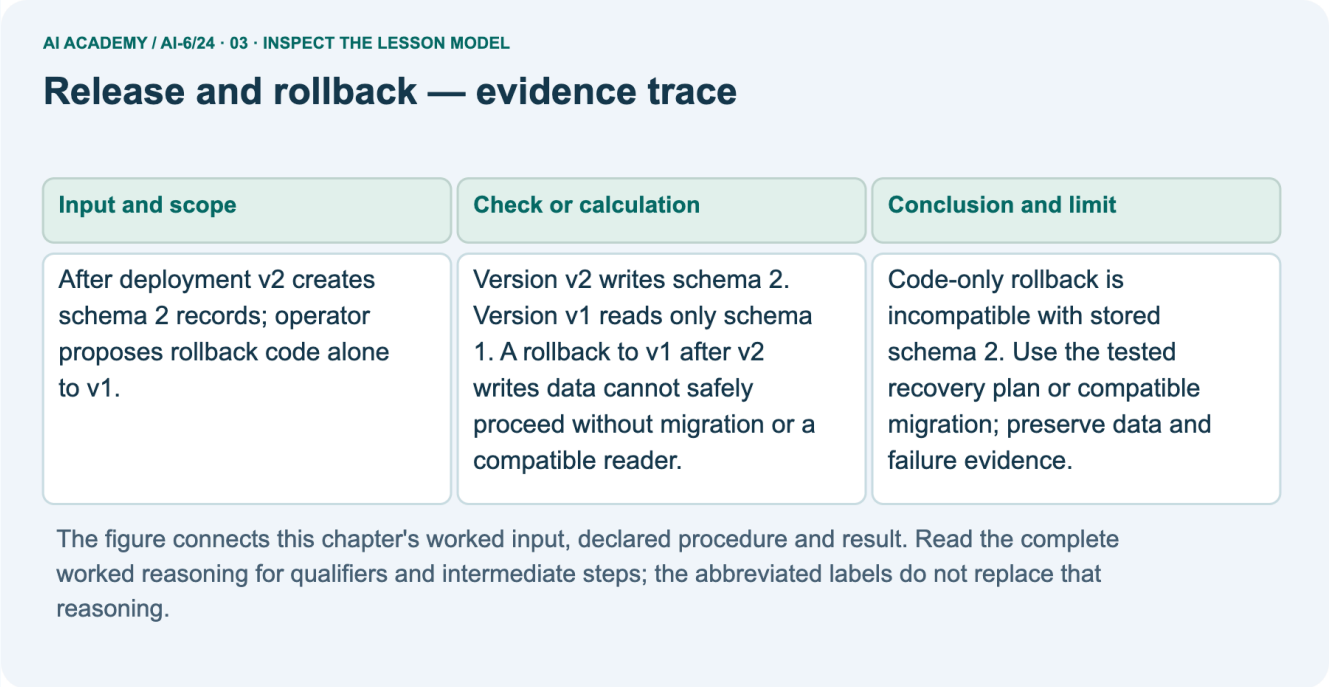
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute quality: Correctness rises from **80%** to 90% on these cases. → Apply privacy gate: The mandatory failure blocks B's release. → Keep recovery ready: Retain the reviewed A version and documented **rollback** procedure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Release and rollback — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Release and **rollback** — **evidence** trace. After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. | Version v2 writes schema 2. Version v1 reads only schema 1.

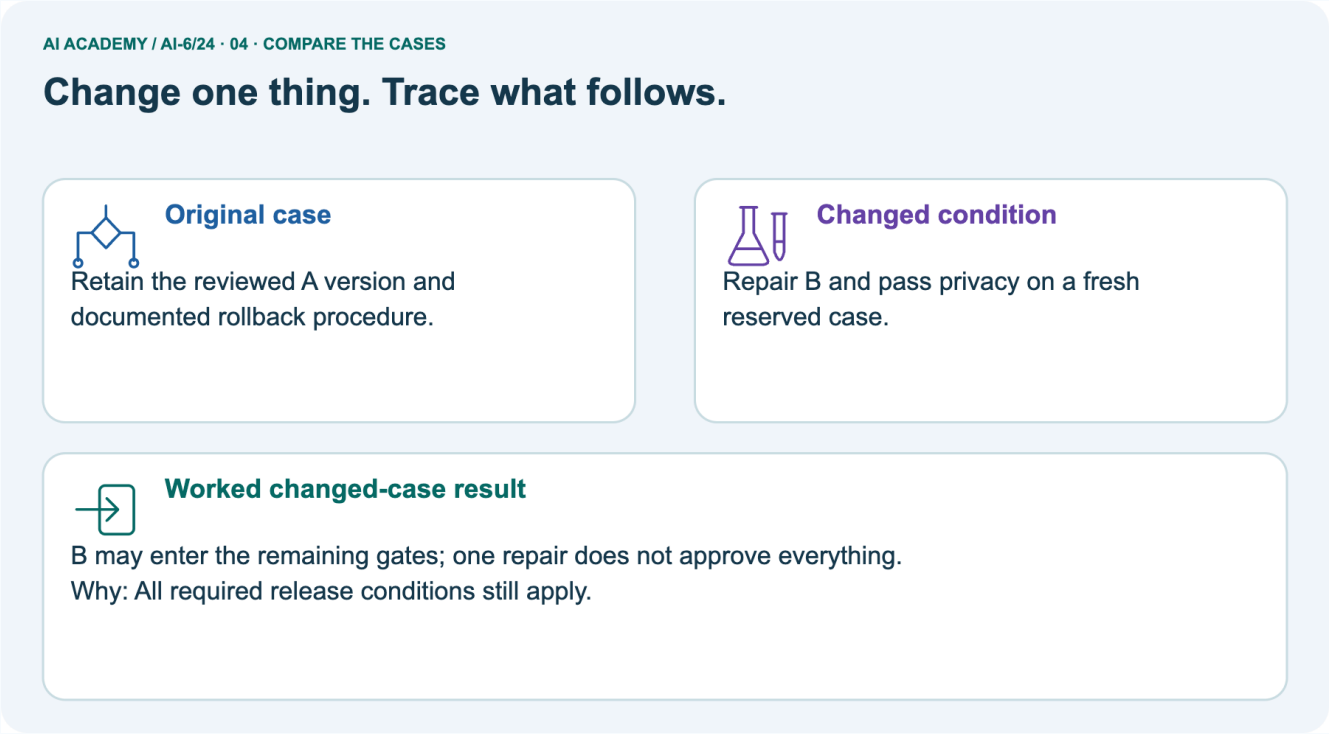
A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader. | Code-only **rollback** is incompatible with stored schema 2.

Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

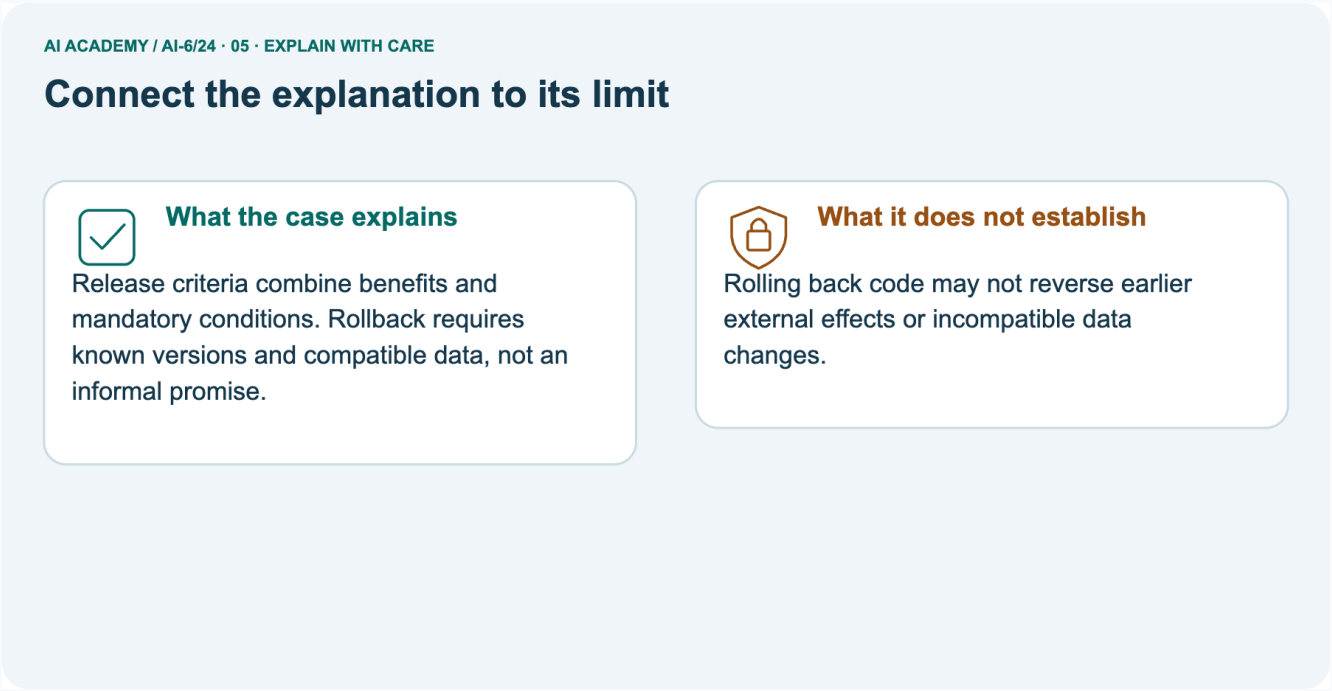
Original case: Retain the reviewed A version and documented **rollback** procedure. Changed condition: Repair B and pass privacy on a fresh reserved case.

Expected result: B may enter the remaining gates; one repair does not approve everything. Reason: All required release conditions still apply.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

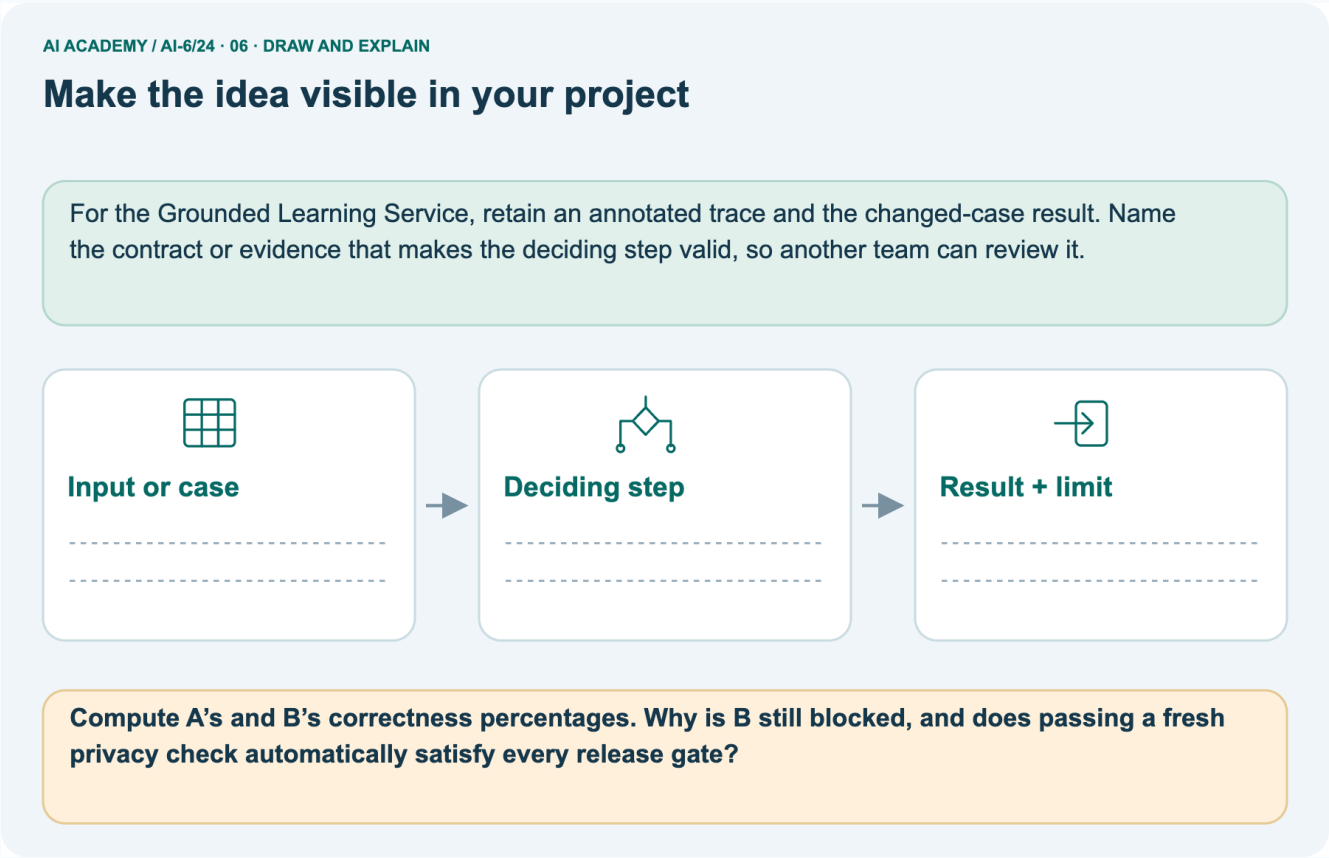
Release criteria combine benefits and mandatory conditions. **Rollback** requires known versions and compatible data, not an informal promise.

Limit: Rolling back code may not reverse earlier external effects or incompatible data changes.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Release and rollback: the case

02. Follow the deciding steps

03. Release and rollback — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.